

Project: Food Remedy API

Ticket: RE-023 – Mobile Application Data Validation and Integrity Best Practices

Author: Gurkirat Singh Chahal

Student ID: 223936693

Date: 17/09/2026

Table of Contents

1. Introduction
2. Food and Product Data Validation
3. Common Data Quality Problems
4. External API Data Validation
5. Client, Backend and Database Validation
6. Handling Uncertain Data and Maintaining Integrity
7. Recommendations for Food Remedy
8. Conclusion
9. References

1. Introduction

To give consumers information about ingredients, nutrition, and allergens, Food Remedy uses food and product data. The accuracy of this data is crucial since application functionalities may show or utilise inaccurate data if it is missing, improperly structured, duplicated, or inconsistent.

When data originates from outside sources like Open Food Facts, this is very important. Ingredients, nutrition, and data-quality fields are only a few of the several product information kinds that are included in its product schema. Nutrition values can have various units and measurement bases, such as per serving or per 100 g.

This study looks on data-integrity and practical validation methods that could be used for Food Remedy prior to its public release. Food-specific validation, typical data-quality issues, external API validation, validation at various application layers, managing information that cannot be reliably validated, and preserving consistency when product information changes are its key areas of focus.

The objective is to offer useful suggestions that minimise inaccurate food information while maintaining the validation procedure appropriate for Food Remedy's current database, backend, and mobile architecture.

2. Food and Product Data Validation

Validation ought to ascertain more than just a field's existence. The right data type, format, unit, and reasonable value should also be present in the information. Semantic validation, which determines whether a value makes sense in its context, and syntactic validation, which determines whether information has the anticipated structure, are both recommended by OWASP.

These ideas can be directly applied to product information for Food Remedy.

Table 1. Recommended Food Data Validation

Data	Example Validation
Barcode	Present and correctly formatted
Product name	Not empty or invalid
Ingredients	Valid text/list structure
Nutrition value	Numeric and within reasonable limits
Nutrition unit	Recognised unit such as g or mg
Nutrition basis	Correctly identified as per 100 g or serving
Allergens	Recognised terminology and consistent with available ingredient information
Update information	Valid timestamp/version where available

For instance, "protein": -15 should be rejected or flagged since a negative protein quantity is meaningless, although "protein": 15 can be structurally OK.

Instead of interpreting all nutrition figures as directly identical, Food Remedy should maintain the distinction made by Open Food Facts regarding whether nutrition information applies per serving or per 100 g.

3. Common Data Quality Problems

Food Remedy may come into information that is inconsistent, redundant, inaccurately presented, or incomplete. Different approaches should be taken to each of these conditions.

Table 2. Common Food-Data Problems

Problem	Example	Recommended Handling
Missing data	Ingredients unavailable	Mark as unavailable
Empty field	<code>ingredients = ""</code>	Treat as missing
Wrong format	Protein contains text	Reject affected value
Invalid value	Negative nutrition quantity	Flag as invalid
Duplicate	Same barcode twice	Reconcile using identifier/version
Unknown unit	Energy value without unit	Avoid calculations
Conflicting information	Ingredients indicate milk but allergen data does not	Mark as uncertain

Outdated information	Newer product record available	Revalidate/update
Missing allergen data	No allergen information	Treat as unknown

Importantly, **missing information should not automatically become a valid value**. Missing sugar information, for example, is not equivalent to 0 g sugar.

Food Remedy could classify information into four simple states:

Valid – required checks have passed.

Incomplete – information is missing but remaining data may still be useful.

Uncertain – information conflicts or cannot be confidently interpreted.

Invalid – important validation requirements have failed.

This prevents incomplete data from appearing more certain than it actually is.

4. External API Data Validation

A successful API request only confirms that information was returned. It does not guarantee that every field is complete, valid or suitable for Food Remedy.

A suitable validation process would be:

External API → Schema Check → Field Validation → Normalisation → Consistency Check → Approved Data

Food Remedy should check:

- whether the API response can be parsed;
- expected fields and data types;
- product identifiers;
- nutrition values, units and measurement basis;
- ingredient and allergen information;
- duplicate records;
- conflicting important fields; and
- freshness/version information where available.

This is particularly important with external APIs because schemas can change. Open Food Facts states that new fields can be introduced over time and recommends that API consumers tolerate additional fields rather than failing when they appear. Breaking field changes are associated with API-version changes.

Food Remedy should therefore validate the fields it actually depends on while safely ignoring unsupported additional fields.

5. Client, Backend and Database Validation

Validation should occur at several layers rather than relying entirely on the mobile application.

Table 3. Validation Approach Comparison

Area	Client/Mobile	Backend/API	Database
Immediate feedback	Strong	Moderate	Weak
Basic format checks	Strong	Strong	Moderate
External API validation	Limited	Strong	Weak
Nutrition validation	Basic	Strong	Basic
Allergen logic	Display checks	Strong	Store validated state
Business rules	Limited	Strong	Limited
Duplicate prevention	Preliminary	Strong	Strong
Structural integrity	Limited	Strong	Strong
Main purpose	Early feedback	Authoritative validation	Final integrity protection

The primary product-data validation should be handled by the backend or API. Information obtained from a client or outside source should not always be trusted, but client-side checks are still helpful for quick feedback and avoiding blatant errors.

OWASP particularly lists external feeds from suppliers and other parties as potentially untrusted inputs and advises checking untrusted data as soon as feasible.

An additional degree of security can be offered by Firestore. Security rules have the ability to assess incoming document data and, depending on predetermined criteria, permit or prohibit writes.

Thus, the model of choice is:

Database integrity controls → Backend/API validation → Mobile checks

6. Handling Uncertain Data and Maintaining Integrity

Food Remedy should refrain from making a firm claim when the information at hand does not support it.

This is particularly crucial when it comes to allergies. When certain allergies are found in Australia and New Zealand, FSANZ mandates that they be declared using the necessary names. For relevant packaged items, current regulations also call for packaging and a separate allergy summary statement.

Consequently, in the event that Food Remedy is provided with inaccurate or contradictory allergy information:

Choose:

Information about allergens could not be confirmed.

Refrain from:

There were no allergies found.

unless that judgement is genuinely supported by the information at hand.

After information has been first verified, data integrity must also be maintained.

Table 4. Data Integrity Practices

Practice	Food Remedy Application
Stable identifier	Use barcode/product identifier
Timestamp	Record when information changed
Source	Record where information originated
Version	Identify dataset/schema version
Duplicate checking	Prevent unintended duplicate products
Revalidation	Validate changed information again
Validation status	Store valid/incomplete/uncertain/invalid
Logging	Record important validation failures
Quarantine	Prevent critically invalid data reaching users

Instead of supposing that information is always up to date, Open Food Facts product metadata supports the idea of tracking product changes by including modification and update information.

7. Recommendations for Food Remedy

Food Remedy should incorporate validation into the entire product-data flow instead of depending just on checks within the mobile interface, according to the report.

Table 5. Recommended Actions

Priority	Recommendation
High	Validate external product data before use
High	Make Backend/API validation authoritative
High	Use valid, incomplete, uncertain and invalid states
High	Never convert missing nutrition information to zero
High	Treat missing allergen information as unknown

High	Validate nutrition units and measurement basis
High	Revalidate information when products change
Medium	Normalise allergen terminology while retaining source information
Medium	Detect duplicates using stable identifiers
Medium	Use Firestore/database integrity controls
Medium	Log significant validation failures

The suggested Food Remedy flow is:

External Information → Database → Mobile Application → Backend Validation → Quality Status

Information should only be considered validated if it passes the required tests. Although incomplete information may still be helpful, it should be made clear in the application rather than being presented as confirmed.

8. Conclusion

Before Food Remedy is released to the public, accurate food, nutrition, and allergen information is crucial. Even if product information is successfully retrieved from an external source, it may not be comprehensive, properly organised, or appropriate for application features.

Product identities, ingredients, nutrition values and units, allergies, duplicates, and connections between related variables should all be verified by Food Remedy. Information that is partial, erroneous, or missing should be identified rather than being taken for granted.

It is advised to use a multi-layer strategy. The backend/API should handle the primary product-data validation, the mobile application should offer basic validation and secure presentation, and Firestore/database controls should offer extra integrity protection.

Missing nutrition and allergen information should be handled with extra caution. Unavailable allergen information should be regarded as unknown rather than allergen-free, and missing nutrition data shouldn't immediately become zero. These procedures would lessen the possibility that consumers of Food Remedy would receive inaccurate product information.

References

OWASP Foundation. (2026). Input Validation Cheat Sheet. OWASP Cheat Sheet Series.
[https://cheatsheetseries.owasp.org/cheatsheets/Input Validation Cheat Sheet.html](https://cheatsheetseries.owasp.org/cheatsheets/Input_Validation_Cheat_Sheet.html)

Open Food Facts. (2026). Product Schema. Open Food Facts Documentation.

<https://openfoodfacts.github.io/documentation/docs/Product-Opener/schemas/schemas/product/>

Open Food Facts. (2026). Explain nutrition data. Open Food Facts Documentation.

<https://developer.android.com/docs/quality-guidelines/core-app-quality>

Google. (2026). Writing conditions for Cloud Firestore Security Rules. Firebase

Documentation. <https://firebase.google.com/docs/firestore/security/rules-conditions>